

No. of Functional Features Included in the Solution

Date	15-07-2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Functional Features Overview

This document lists all the functional features included in the Flood Prediction System solution. Each feature describes its purpose, implementation status, and contribution to the overall system.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Input Data Collection	Allows users to enter environmental parameters such as rainfall, temperature, humidity, cloud visibility, and seasonal data for prediction.	User Interface Module	Done	1
2	Data Preprocessing	Cleans and transforms input data into a suitable format for machine learning model prediction.	Data Processing Module	Done	1
3	Flood Prediction Using ML Model	Predicts flood risk based on input parameters using trained machine learning algorithms.	ML Prediction Engine	Done	1
4	Machine Learning Model Integration	Integrates trained models such as Random Forest, Decision Tree, KNN, and XGBoost with the application.	ML Model Module	Done	1
5	Prediction Result Display	Displays flood prediction results and risk levels to users in an understandable format.	Result Visualization Module	Done	1
6	Data Visualization Dashboard	Provides graphical representation of prediction results and environmental factors using charts.	Dashboard Module	Done	1
7	Historical Dataset Management	Stores and manages historical flood and weather datasets used for model training and analysis.	Database / Dataset Module	Done	1
8	Web Application Interface	Provides a user-friendly web interface for accessing flood prediction services through a browser.	Flask Web Application	Done	1

Feature Summary

Metric	Count / Value
--------	---------------

Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Input form, Prediction dashboard, Result display
2	Backend / Logic	3	Data preprocessing, ML prediction, Model integration
3	Database / Storage	1	Historical flood dataset storage
4	API / Integration	1	Flask application integration with ML model
5	Security / Authentication	1	Input validation and secure application handling

Project: Flood Prediction System Using Machine Learning
Technology Stack: Python, Flask, Machine Learning, Pandas, Scikit-learn, XGBoost, HTML, CSS, JavaScript